Data Input
Raw E-Commerce Data
(2014-2018, 48 months)
~10,000 transactions
Data Preprocessing
Missing Value Treatment
(Postal Code → Unknown)
Outlier Detection
(Z-score method, threshold=3)
Data Validation
(Type checking, duplicates)
Monthly Aggregation
(Daily → Monthly sales)
Feature Engineering (43 Features)
Lag Features (12)
Lags: 1, 3, 6, 12 months
Sales, Orders, Profit
Rolling Statistics (12)
Mean, Std, Min, Max
Windows: 3, 6, 12
Date Features (7)
Month, Quarter, Year
Sin/Cos encoding
Growth Metrics (6)
MoM, YoY growth
Momentum, Volatility
Statistical Features (6)
Z-score, Percentile
Deviation from mean
Train/Test Split
Train: 36 months (Jan 2015 - Dec 2017)
Test: 12 months (Jan 2018 - Dec 2018)
Parallel Model Development
Model 1: Baseline
Linear Regression
Trend + Seasonal
Model 2: Statistical
Facebook Prophet
Additive decomposition
MCMC sampling
Model 3: Deep Learning
LSTM Network
50 units, 100 epochs
Sequence length: 12
Model 5: Gradient Boosting
XGBoost
100 trees, depth=4
L1/L2 regularization
60%
40%
Model 4: Weighted Ensemble
60% Prophet + 40% LSTM
Linear combination
Test Set Evaluation
Metrics: MAPE, R², MAE, RMSE
Direction Accuracy
Comprehensive Validation
Cross-Validation
24-fold Walk-Forward
Expanding window
Statistical Tests
Paired t-test
Friedman test
α = 0.05
Diagnostic Tests
Normality (Shapiro-Wilk)
Bias (One-sample t-test)
Autocorrelation
Feature Importance
SHAP values
Gain-based importance
Model Performance Results
Linear: MAPE 25.3%
LSTM: MAPE 30.3%
Prophet: MAPE 19.6%
R² 0.865
Ensemble: MAPE 15.2%
R² 0.826
XGBoost: MAPE 11.6%
R² 0.856 ★ BEST
Production Model
XGBoost Deployment
Monitoring Framework
Retraining Pipeline